WRAYSBURY MATERIALS LABORATORY

Independent materials testing & failure analysis · Wraysbury, Berkshire TW19 5NB

FAILURE ANALYSIS REPORT — REPORT No. WML-2025-0042

Client: Verdant Energy Systems Limited · Subject: CH-280 LFP cells, String B, Brackmoor Project · Date: 24 January 2025

1. Instructions and scope

1.1 Verdant Energy Systems Limited instructed Wraysbury Materials Laboratory to investigate the cause of abnormal electrical and thermal behaviour observed in module string “String B” of CH-280 lithium iron phosphate (LFP) cells supplied by Calderhall Cells Limited and installed at the Brackmoor energy-storage facility. We were provided with 24 representative cells removed from nine affected modules, together with the commissioning test logs and battery-management system (BMS) data.

2. Methodology

2.1 We carried out: (a) incoming inspection and electrical characterisation (capacity, DC internal resistance, open-circuit voltage); (b) computed tomography (CT) of intact cells; (c) controlled discharge and teardown of a sub-sample in an inert atmosphere; (d) scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) of electrode and separator material; and (e) review of the supplied BMS and commissioning data.

3. Findings

3.1 **Capacity and resistance.** 19 of the 24 cells delivered less than 90% of rated capacity; 11 cells showed DC internal resistance more than 35% above the population mean, indicating localised degradation.

3.2 **Cathode coating.** SEM imaging revealed delamination of the cathode coating from the aluminium current collector over substantial areas in 14 cells, with coating-thickness variation well outside a controlled process window. This is consistent with a process excursion during electrode calendering or drying.

3.3 **Separator contamination.** EDS detected metallic and ceramic particulate inclusions embedded in the separator of 9 cells. Particulates of this nature provide nucleation sites for dendrite growth and localised heating.

3.4 **Lithium plating.** Teardown of three cells showed grey metallic deposits on the anode surface characteristic of lithium plating. Plating of this kind is associated with non-uniform electrode loading and is a recognised precursor to internal short-circuit and thermal runaway.

4. Discussion

4.1 The defects observed are manufacturing defects originating in cell production. They are not consistent with damage caused by the manner of installation, integration or thermal management of the modules. The localised heating recorded by the BMS, and the two thermal-protection interlock trips, are explained by the internal defects identified above and would, if the cells continued in service, present a credible risk of thermal runaway.

4.2 In our opinion the affected cells were not fit for use in a stationary energy-storage installation and should not be energised. As the cells derive from a single production campaign that also supplied Strings

A and C, we recommend screening and accelerated-ageing of representative cells from those strings before they are placed in service.

5. Conclusion

5.1 On the balance of the evidence, String B failed by reason of manufacturing defects in the cells — cathode-coating delamination, inconsistent electrode loading, separator particulate contamination and lithium plating — and not by reason of any act or omission of Verdant.

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This report is prepared for the instructing client. Where it is to be relied upon as expert evidence in proceedings, it will be reissued in CPR Part 35 compliant form with the requisite declaration.